

Digital Forensics Case Handling, Autopsy and Sleuth Kit Analysis

OJE DOMINION MAYOWA

FWSD2511507

Computer and Digital Forensics

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LAB OVERVIEW

This lab focused on the **forensic examination and recovery of digital evidence** from a disk image while preserving its integrity. We learned to calculate **MD5 and SHA-256 hashes** to authenticate evidence, create and analyze a forensic case using **Autopsy**, and examine disk structures using **The Sleuth Kit (TSK)**.

Key activities included using `mmls` to identify partition offsets, `fsstat` to examine file-system information, and `fls` to locate files and deleted files. We used `istat` to examine file metadata and timestamps, `icat` to recover specific files such as **Income.xls**, and `blkcat` to inspect individual data blocks. We also used `tsk_recover` to recover multiple files from the forensic image.

Finally, recovered files were verified using **MD5/SHA-256 hash comparisons**, ensuring that the recovered evidence remained accurate and unchanged. Overall, the lab demonstrated the process of **preserving, analyzing, recovering, and validating digital evidence in a forensic investigation**.

OBJECTIVES

- **Preserve Evidence:** Calculate and verify MD5 and SHA-256 hashes to maintain data integrity.
- **Perform Forensic Analysis:** Analyze the disk image using Autopsy and identify relevant/deleted files.
- **Analyze Disk Structure:** Use TSK tools to examine partitions, file systems, sectors, and directories.
- **Examine Metadata:** Analyze file metadata, timestamps, and data block allocation using `istat`.
- **Recover Evidence:** Extract deleted files and data using `icat`, `blkcat`, and `tsk_recover`.
- **Validate Recovery:** Compare hashes of recovered files to verify data authenticity.
- **Prepare Forensic Report:** Document procedures, findings, errors, and evidence in a reproducible report.

TOOLS

- Terminal/Terminator
- Autopsy
- `img_stat`
- `mmls`, `ffsstat`, `fls`
- `istat`, `icat`
- `blkcat`, `blkls`
- `tsk_recover`
- `md5sum`, `sha256sum`.
- Sleuth Kit

METHODOLOGY

The authorised training forensic image was downloaded firstly to begin the lab



Fig 1.1 Downloaded Forensic Image

Next, we create a dedicated working directory named
FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION

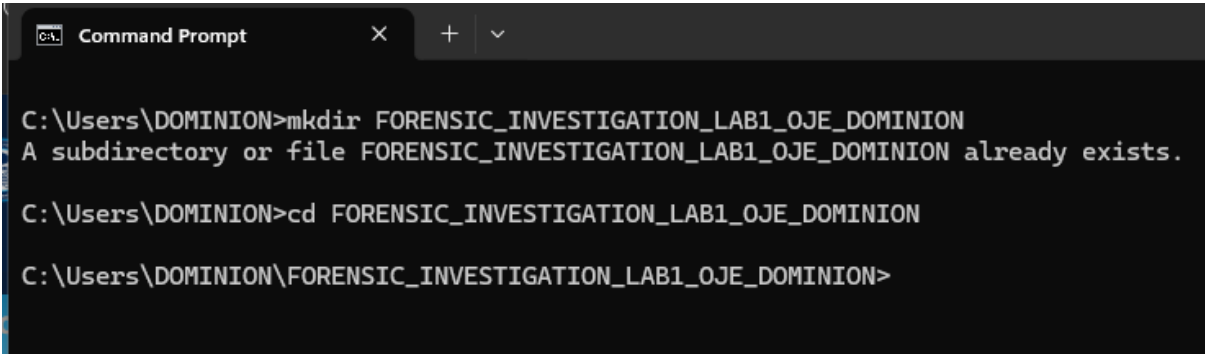


Fig 1.2 Lab Environment Created

PRE-ANALYSIS VERIFICATION TABLE

Evidence source	Ch01InChap01.dd (Authorized source image from the IC DFA portal)
File size	1440KB
Storage location	C:\Users\DOMINION\FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION\Evidence
MD5 hash	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA-256 hash	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

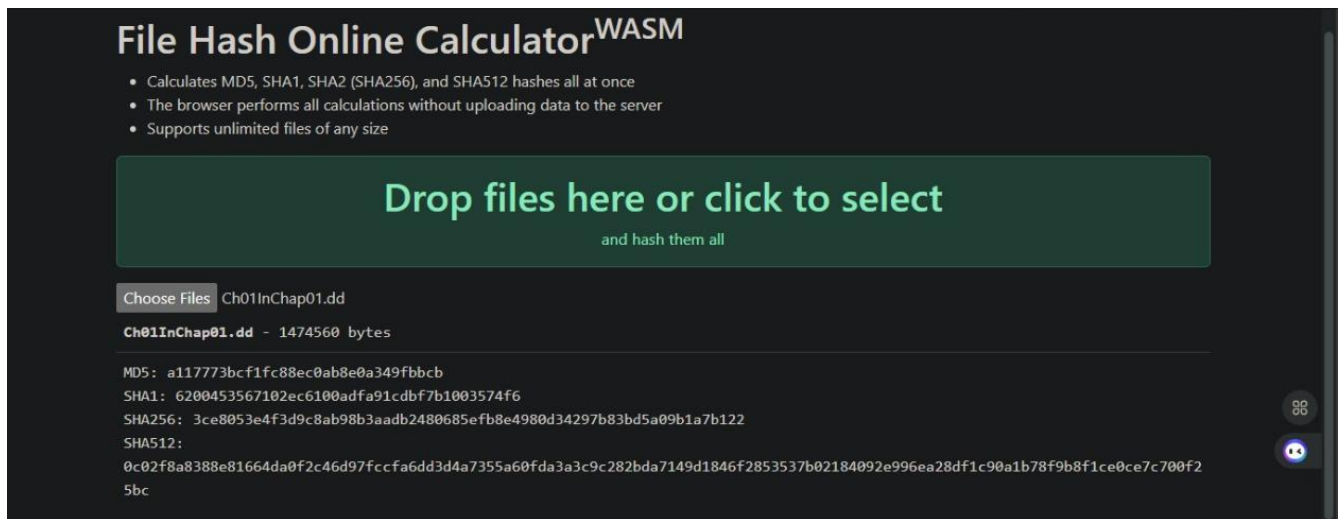


FIG 1: Online checking of the different Encryption of the source file.

PART A

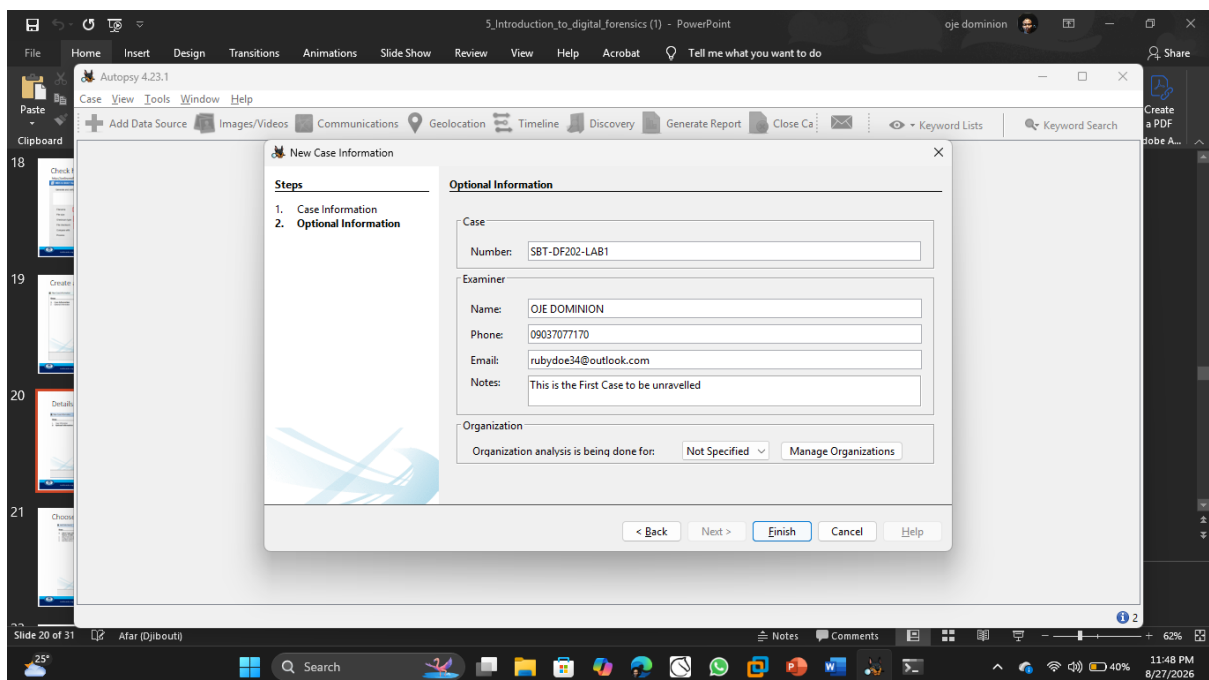


Fig 2: The First Case Created on Autopsy

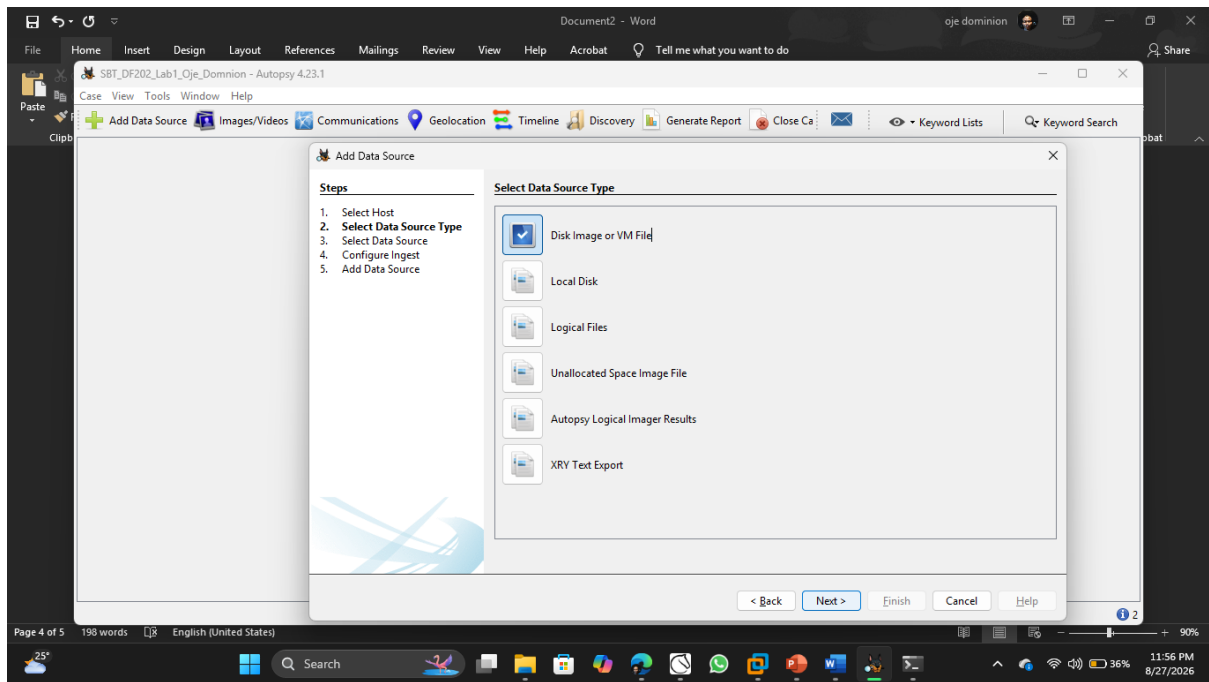


Fig 3: Setting up the Case Lab on Autopsy

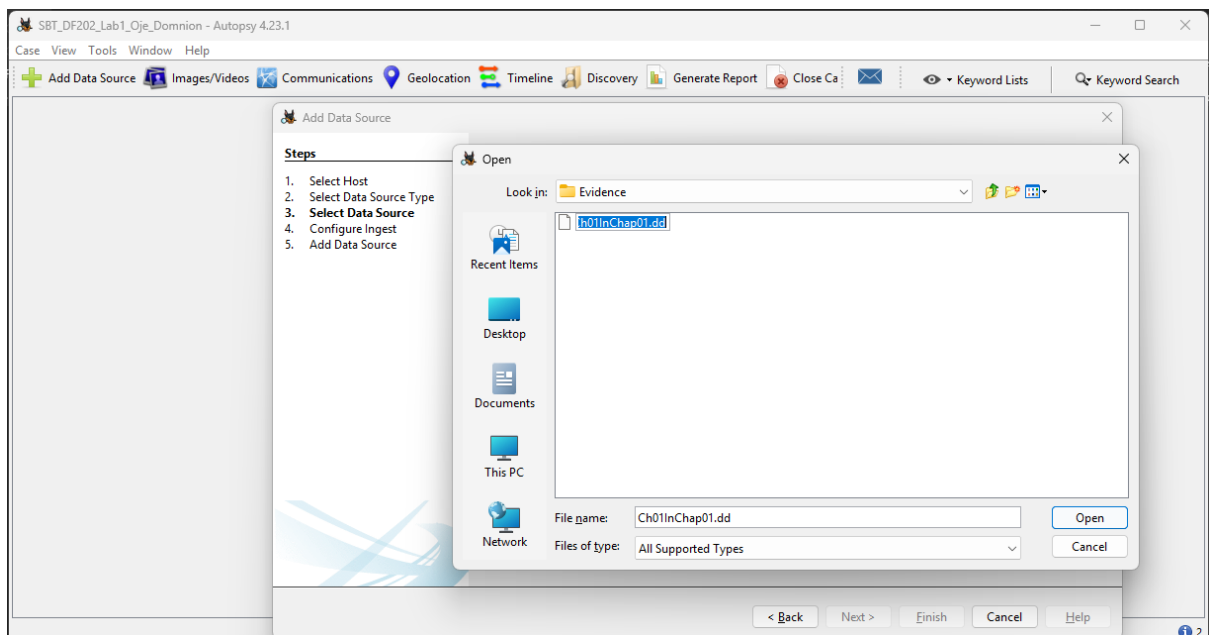


Fig 4: Selecting the Source Image or File

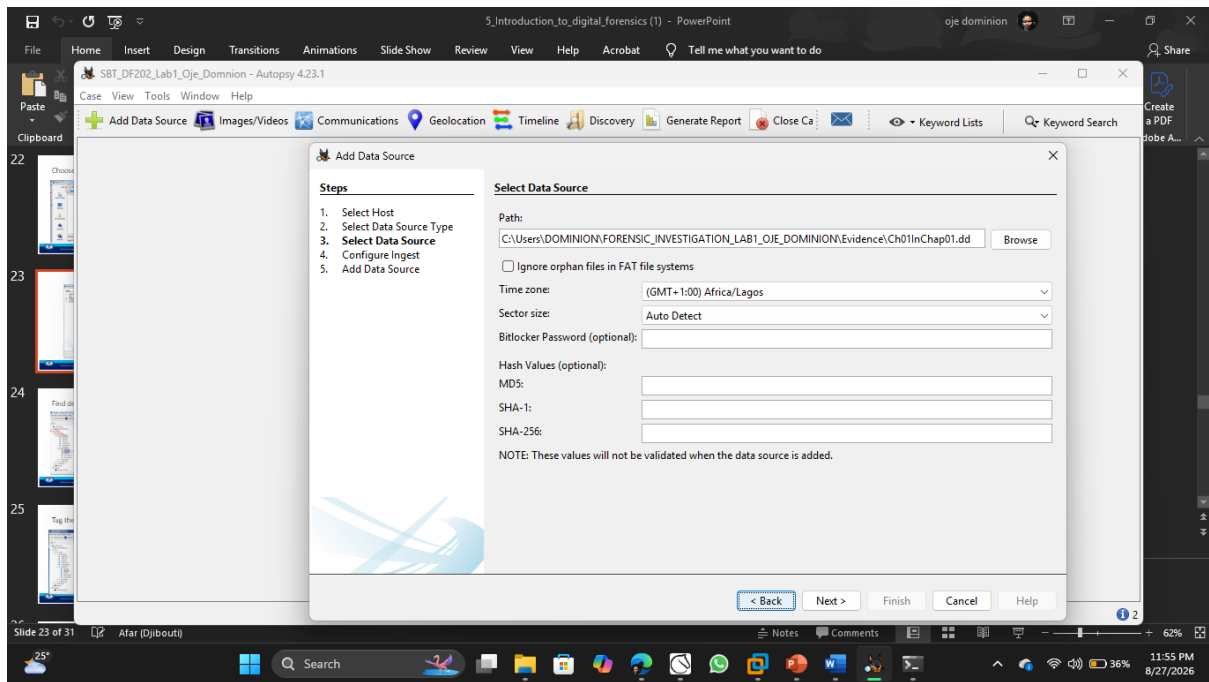


Fig 5: The First Case Created on Autopsy

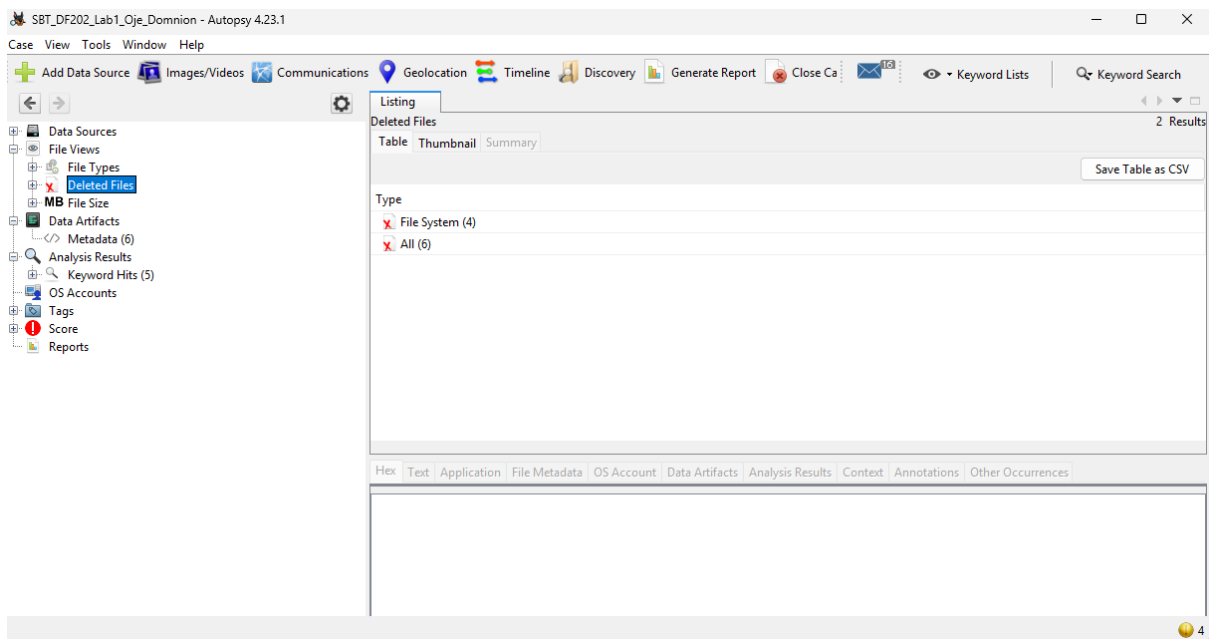


Fig 6: List of files in the system

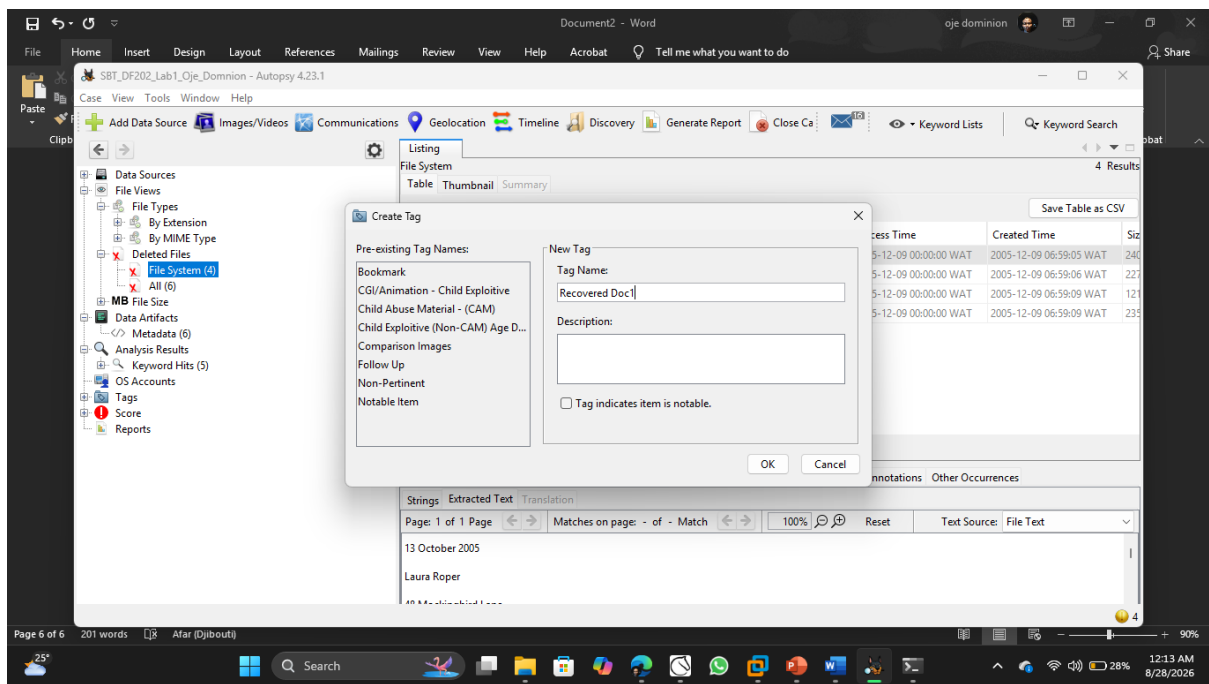


Fig 7: Tagged Recovered File

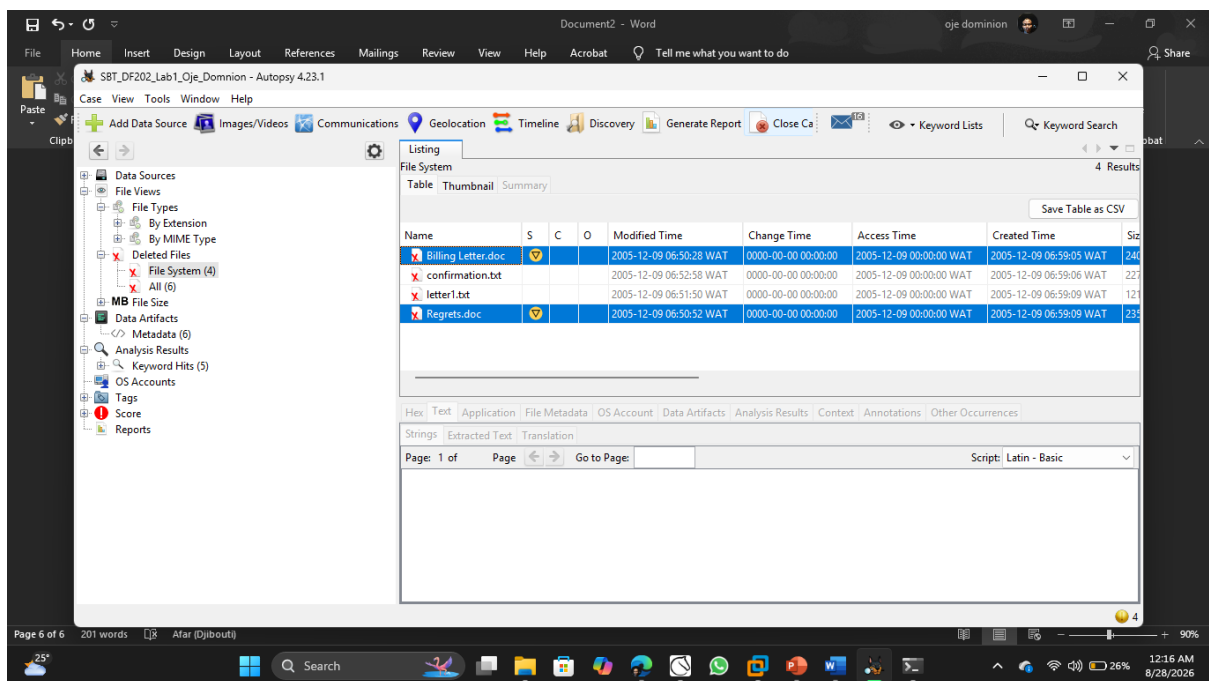


Fig 8: List of Tagged files in the system

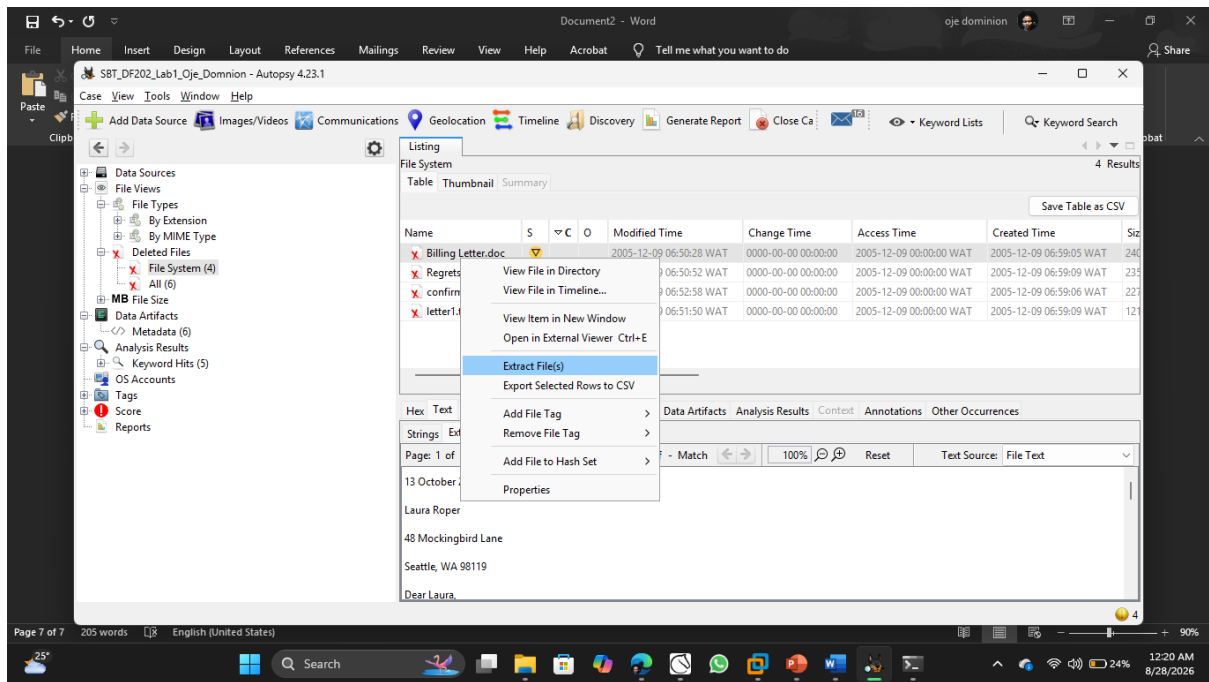


Fig 9: Extracting of files in the system for Recovery

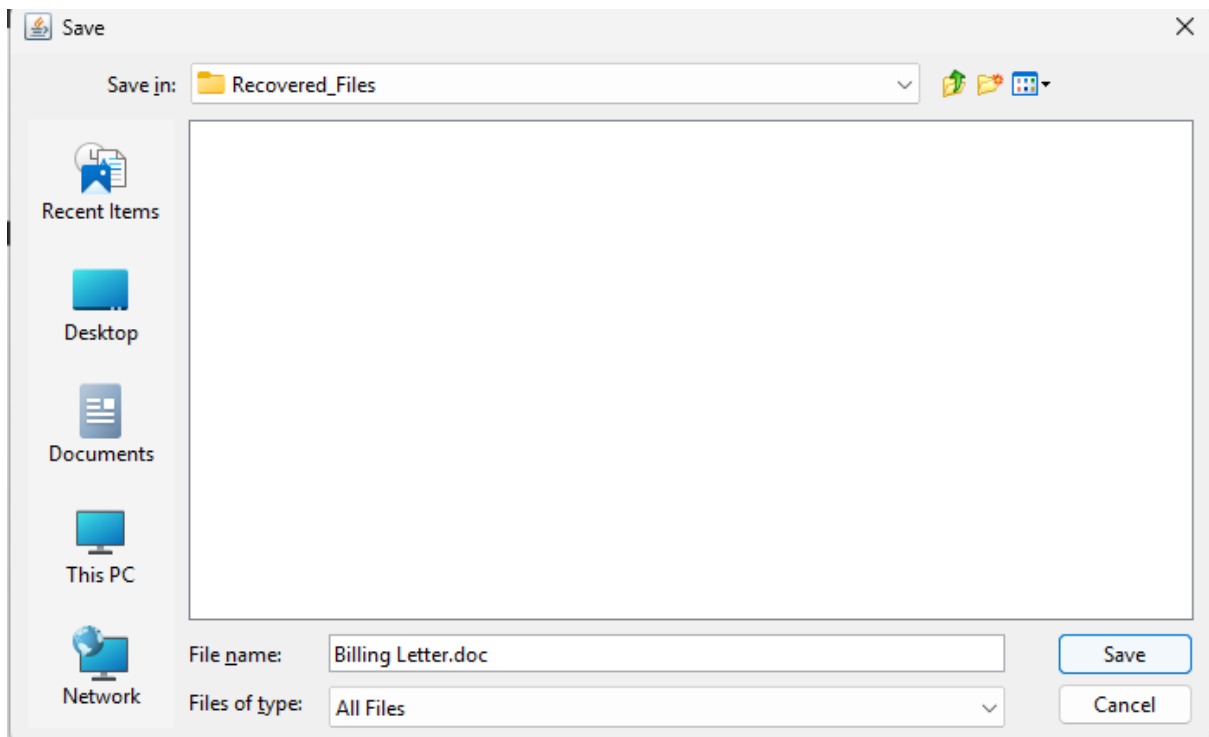


Fig 10: Recovered Files Pane

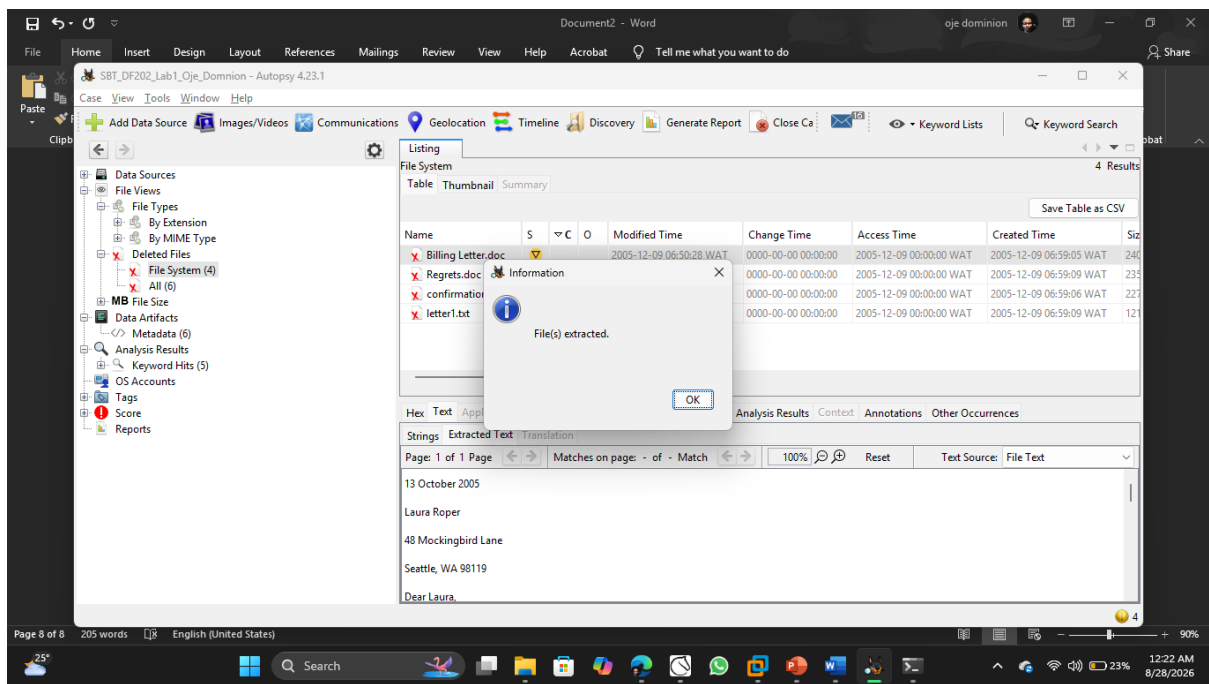


Fig 11: Files Recovered Successfully

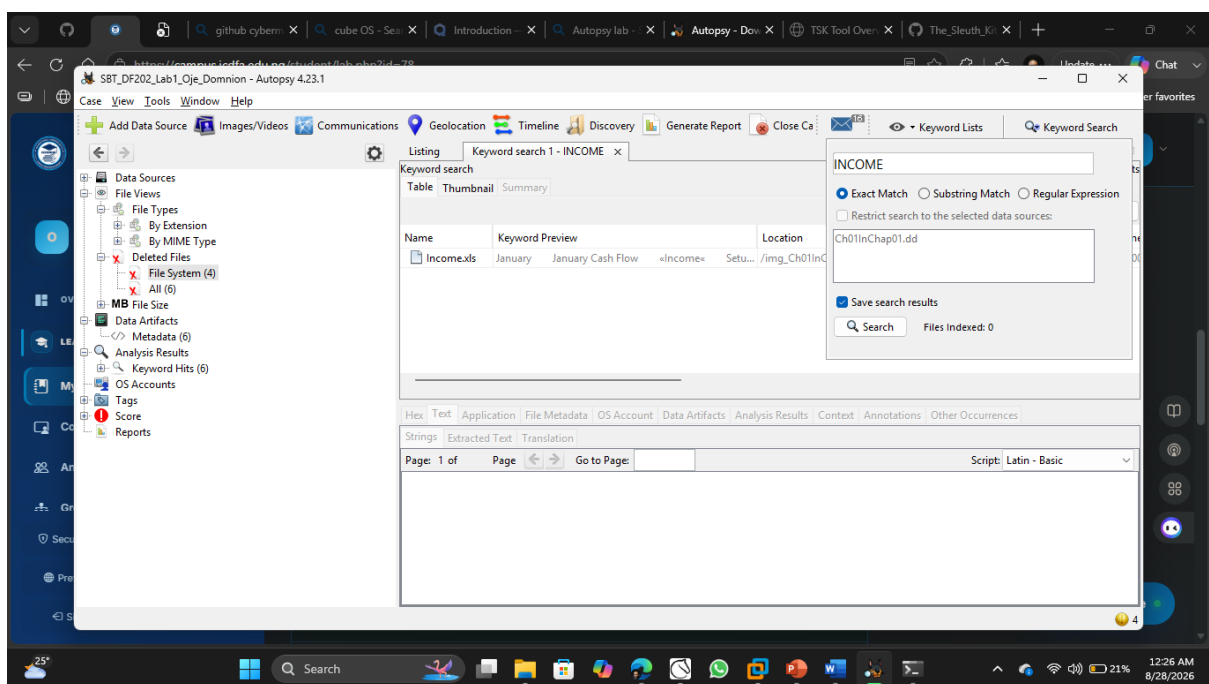


Fig 12: Keyword Search Performed on Autopsy

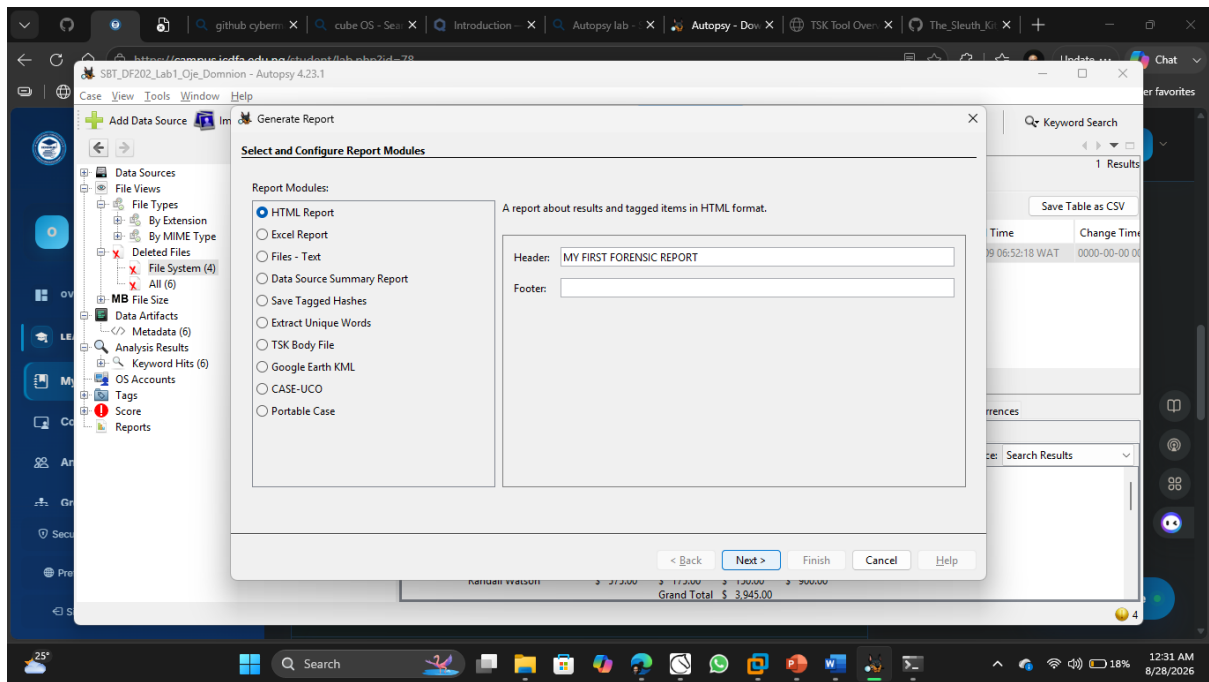


Fig 13: Generating Report from the case

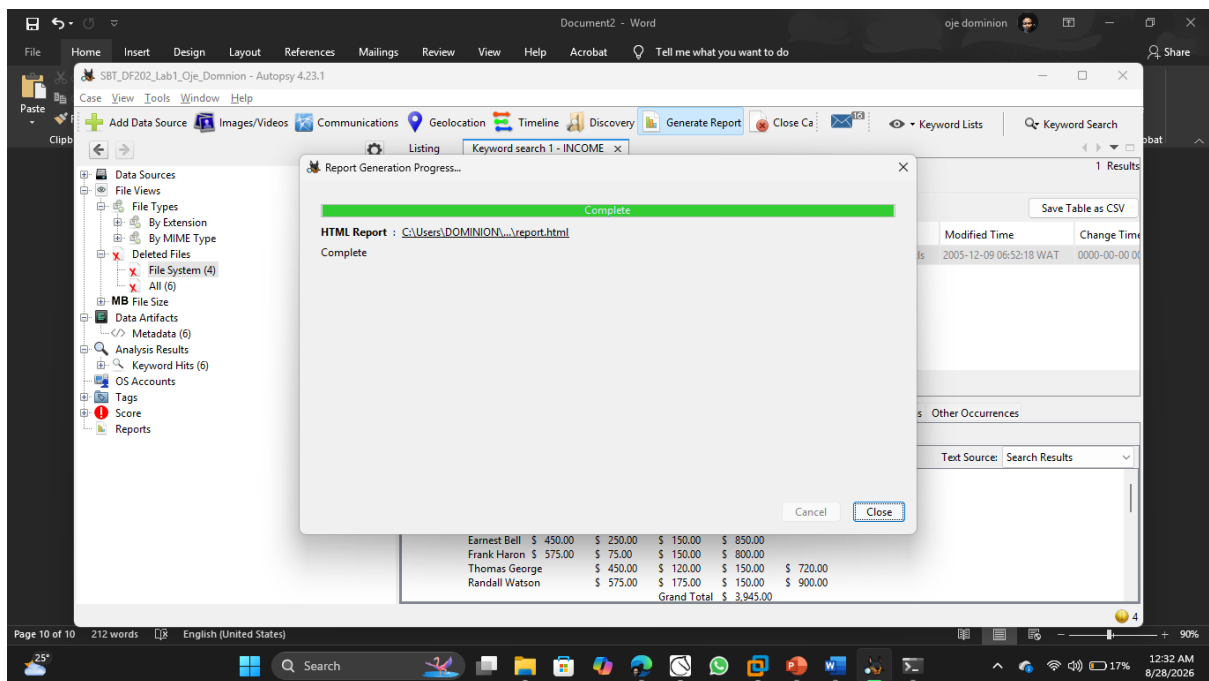


Fig 14: Reports Generated to be viewed as an HTML File

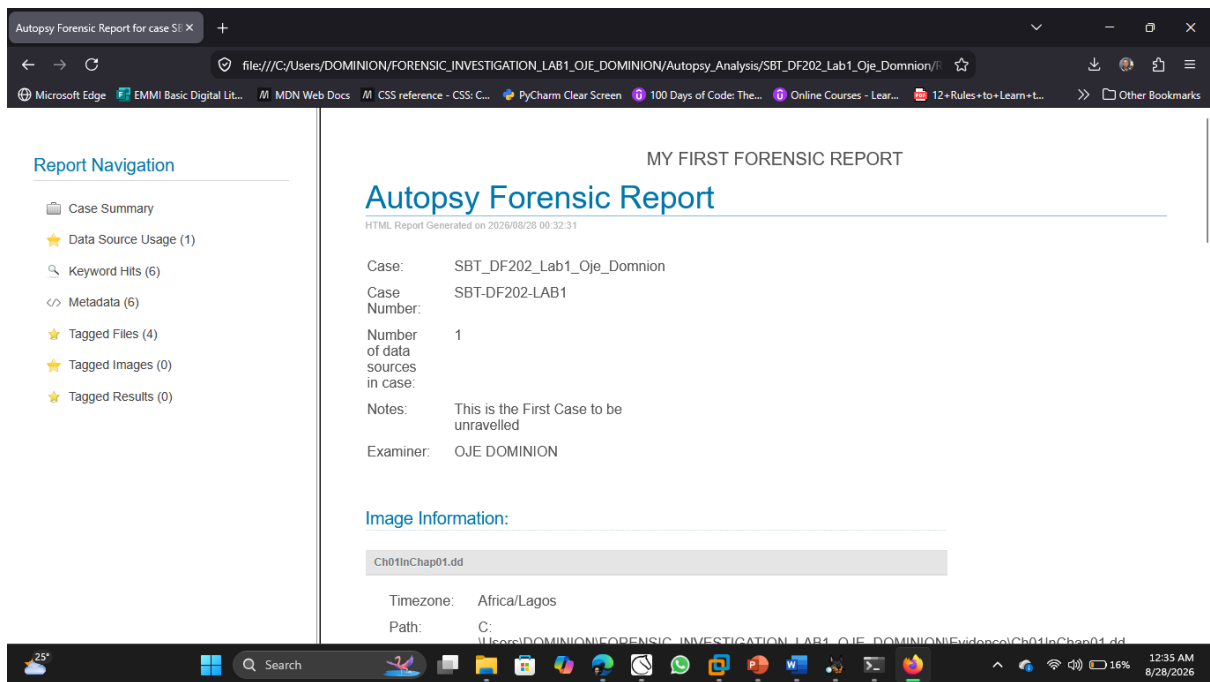


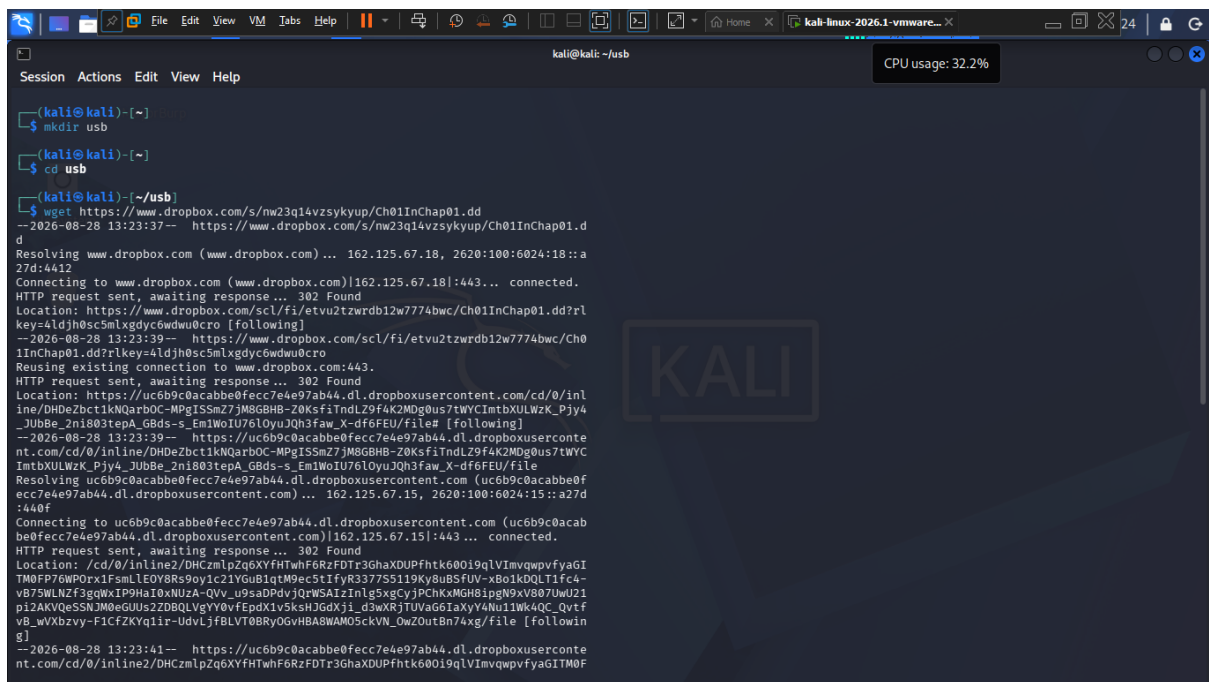
Fig 15: HTML view of the Generated Forensic Summary Report

FORENSIC ANALYSIS SUMMARY TABLE

case name	SBT_DF202_Lab1_Oje_Domnion
evidence source	C:\Users\DOMINION\FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION\Evidence\Ch01InChap01.dd
deleted-file findings	/Ch01InChap01.dd/Billing Letter.doc /Ch01InChap01.dd/Regrets.doc / Ch01InChap01.dd/Confirmation.txt / Ch01InChap01.dd/letter.txt
recovered files	/img_Ch01InChap01.dd/Income.xls /img_Ch01InChap01.dd/Billing Letter.doc /img_Ch01InChap01.dd/Regrets.doc
keyword-search results	/img_Ch01InChap01.dd/Income.xls
reporting evidence	file:///C:/Users/DOMINION/FORENSIC_INVESTIGATION_LAB1_OJE_DOMINION/Autopsy_Analysis/SBT_DF202_Lab1_Oje_Domnion/Reports/SBT_DF202_Lab1_Oje_Domnion HTML Report 08-28-2026-00-32-31/report.html

Part B - Sleuth Kit Analysis

The first step is to create a usb directory where the evidence image would be stored and then to download the evidence image into the directory using wget.



```
kali@kali: ~ -usb
Session Actions Edit View Help
CPU usage: 32.2%

(kali@kali)~-[~]
$ mkdir usb

(kali@kali)~-[~]
$ cd usb

(kali@kali)~-[~/usb]
$ wget https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
--2026-08-28 13:23:37-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.d
d
Resolving www.dropbox.com (www.dropbox.com) ... 162.125.67.18, 2620:100:6024:18::a
27d:4412
Connecting to www.dropbox.com (www.dropbox.com)|162.125.67.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rl
key=4ldjh0sc5mlxgdy6wdwu0cro [following]
--2026-08-28 13:23:39-- https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch0
1InChap01.dd?rlkey=4ldjh0sc5mlxgdy6wdwu0cro
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxusercontent.com/cd/0/inl
ine/DHDeZbct1kNqarBOC-MPgISmZ7jM8GBHB-Z0KsfiTndLZ9f4K2MDg0us7tWYCImtbXULWZK_Pjy4
-JUbBe_2ni803tepA_GBds-s_Em1WoIU76lOyuJQh3faw_X-df6FEU/file# [following]
--2026-08-28 13:23:39-- https://uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxuserconte
nt.com/cd/0/inline/DHDeZbct1kNqarBOC-MPgISmZ7jM8GBHB-Z0KsfiTndLZ9f4K2MDg0us7tWYCI
mtbXULWZK_Pjy4-JUbBe_2ni803tepA_GBds-s_Em1WoIU76lOyuJQh3faw_X-df6FEU/file
Resolving uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxusercontent.com (uc6b9c0acabbe0f
ecc7e4e97ab44.dl.dropboxusercontent.com) ... 162.125.67.15, 2620:100:6024:15::a27d
:440f
Connecting to uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxusercontent.com (uc6b9c0acab
be0fecc7e4e97ab44.dl.dropboxusercontent.com)|162.125.67.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHCzmlpZq6XYfHTwhF6RzFDTr3GhaXDUPfhtk600i9qLVImvqwpvfyagI
TM0FP76WP0rx1FsmLEOY8Rs9oy1c21YGub1qtM9ec5tIfyR3377S5119Ky8uBSFUV-xBo1kDQLT1fc4-
vB75WLNZf3gqWxIP9HaI0xNUZA-QVv_u9saDPdvjQrWSAIzInlg5xgCyjPChKxMGH8ipgN9xV807UwU21
pi2AKVQeSSNJMOeGUUs2ZDBQLVgYy0vfEpdX1v5ksHJGdXji_d3wXRjTUVaG6IaXyY4Nu11Wk4QC_Qvtf
vB_wVxbzvy-F1CfZKYq1ir-UdvLjfbLVT0BRyOGvHBA8WAM05ckVN_OwZ0utBn74xg/file [followin
g]
--2026-08-28 13:23:41-- https://uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxuserconte
nt.com/cd/0/inline2/DHCzmlpZq6XYfHTwhF6RzFDTr3GhaXDUPfhtk600i9qLVImvqwpvfyagITM0F
P76WP0rx1FsmLEOY8Rs9oy1c21YGub1qtM9ec5tIfyR3377S5119Ky8uBSFUV-xBo1kDQLT1fc4-vB75
WLNZf3gqWxIP9HaI0xNUZA-QVv_u9saDPdvjQrWSAIzInlg5xgCyjPChKxMGH8ipgN9xV807UwU21pi2A
KVQeSSNJMOeGUUs2ZDBQLVgYy0vfEpdX1v5ksHJGdXji_d3wXRjTUVaG6IaXyY4Nu11Wk4QC_QvtfvB_w
Vxbzvy-F1CfZKYq1ir-UdvLjfbLVT0BRyOGvHBA8WAM05ckVN_OwZ0utBn74xg/file
Reusing existing connection to uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxusercontent
.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/octet-stream]
Saving to: 'Ch01InChap01.dd'

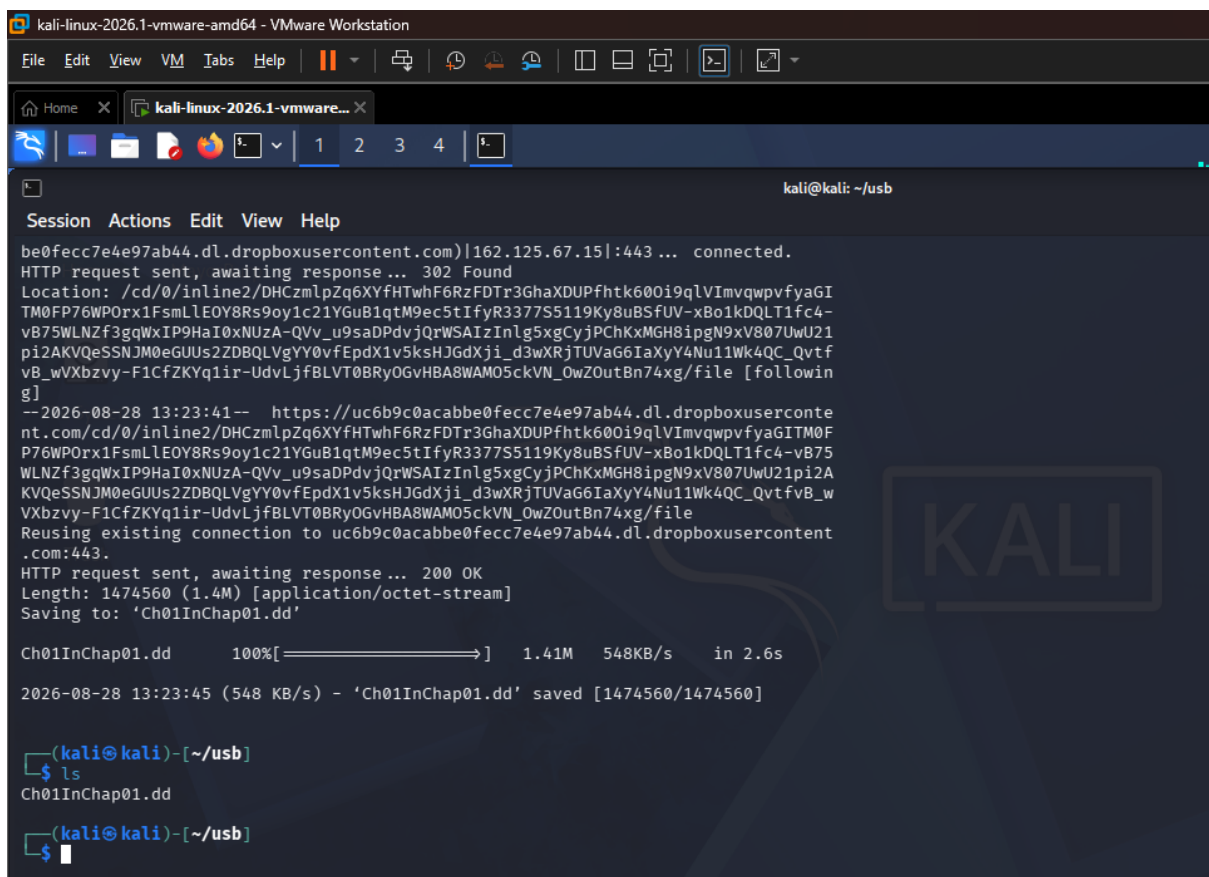
Ch01InChap01.dd      100%[=====] 1.41M  548KB/s   in 2.6s

2026-08-28 13:23:45 (548 KB/s) - 'Ch01InChap01.dd' saved [1474560/1474560]

(kali@kali)~-[~/usb]
$ ls
Ch01InChap01.dd

(kali@kali)~-[~/usb]
$
```

Fig 1.2 Setting up the USB Directory



```
kali@kali: ~ -usb
Session Actions Edit View Help

be0fecc7e4e97ab44.dl.dropboxusercontent.com)|162.125.67.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHCzmlpZq6XYfHTwhF6RzFDTr3GhaXDUPfhtk600i9qLVImvqwpvfyagI
TM0FP76WP0rx1FsmLEOY8Rs9oy1c21YGub1qtM9ec5tIfyR3377S5119Ky8uBSFUV-xBo1kDQLT1fc4-
vB75WLNZf3gqWxIP9HaI0xNUZA-QVv_u9saDPdvjQrWSAIzInlg5xgCyjPChKxMGH8ipgN9xV807UwU21
pi2AKVQeSSNJMOeGUUs2ZDBQLVgYy0vfEpdX1v5ksHJGdXji_d3wXRjTUVaG6IaXyY4Nu11Wk4QC_Qvtf
vB_wVxbzvy-F1CfZKYq1ir-UdvLjfbLVT0BRyOGvHBA8WAM05ckVN_OwZ0utBn74xg/file [followin
g]
--2026-08-28 13:23:41-- https://uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxuserconte
nt.com/cd/0/inline2/DHCzmlpZq6XYfHTwhF6RzFDTr3GhaXDUPfhtk600i9qLVImvqwpvfyagITM0F
P76WP0rx1FsmLEOY8Rs9oy1c21YGub1qtM9ec5tIfyR3377S5119Ky8uBSFUV-xBo1kDQLT1fc4-vB75
WLNZf3gqWxIP9HaI0xNUZA-QVv_u9saDPdvjQrWSAIzInlg5xgCyjPChKxMGH8ipgN9xV807UwU21pi2A
KVQeSSNJMOeGUUs2ZDBQLVgYy0vfEpdX1v5ksHJGdXji_d3wXRjTUVaG6IaXyY4Nu11Wk4QC_QvtfvB_w
Vxbzvy-F1CfZKYq1ir-UdvLjfbLVT0BRyOGvHBA8WAM05ckVN_OwZ0utBn74xg/file
Reusing existing connection to uc6b9c0acabbe0fecc7e4e97ab44.dl.dropboxusercontent
.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/octet-stream]
Saving to: 'Ch01InChap01.dd'

Ch01InChap01.dd      100%[=====] 1.41M  548KB/s   in 2.6s

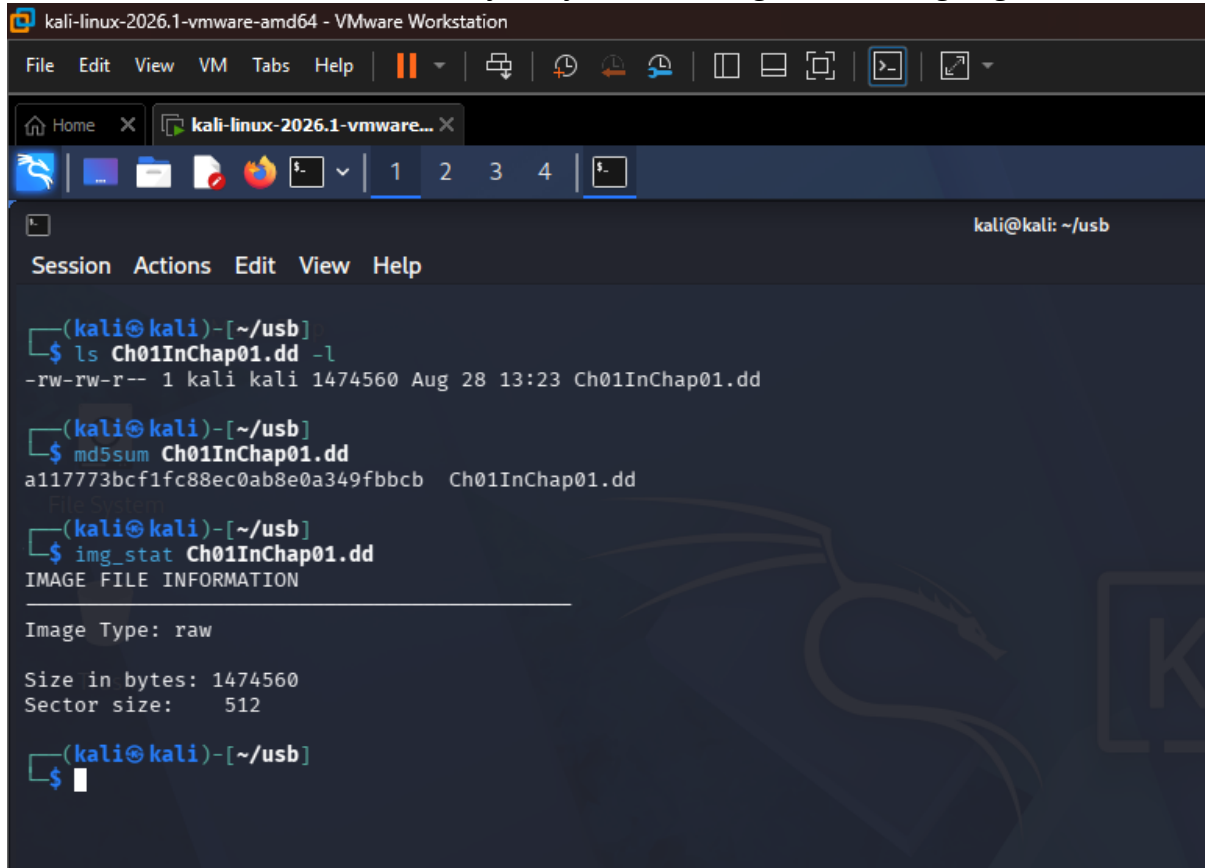
2026-08-28 13:23:45 (548 KB/s) - 'Ch01InChap01.dd' saved [1474560/1474560]

(kali@kali)~-[~/usb]
$ ls
Ch01InChap01.dd

(kali@kali)~-[~/usb]
$
```

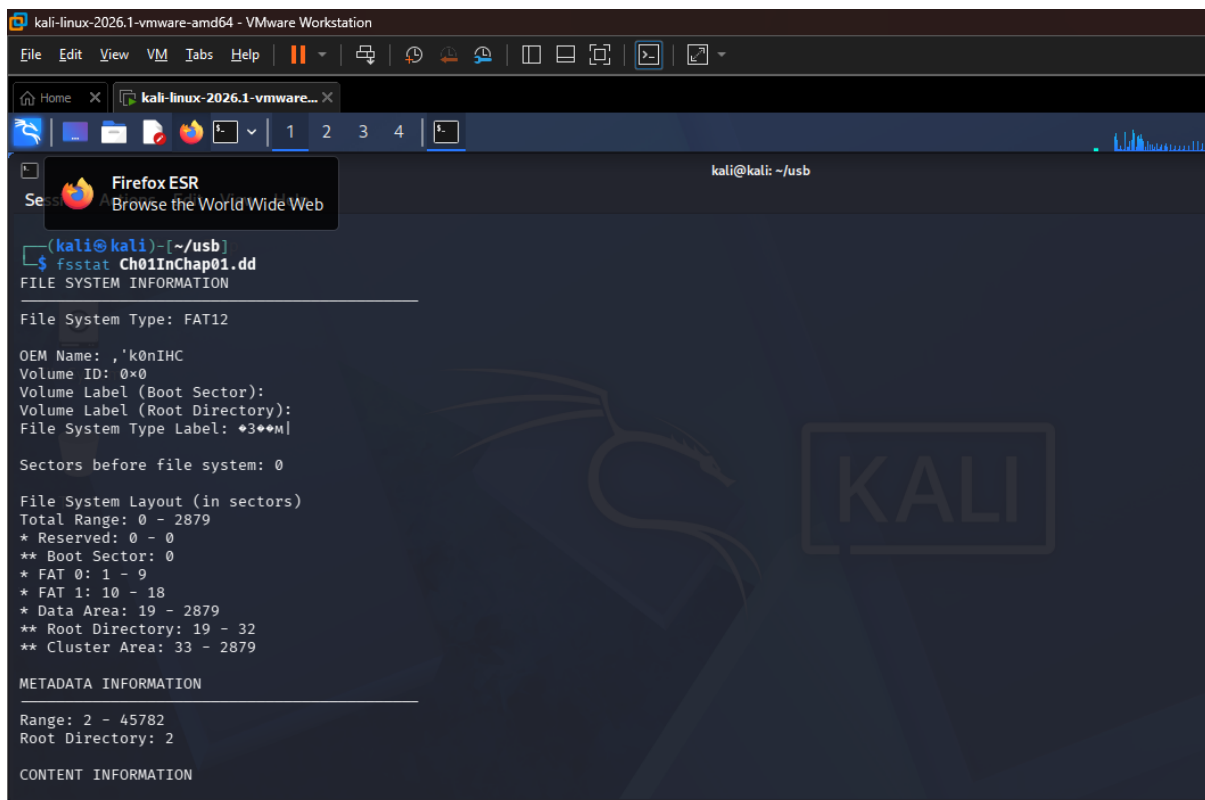
Fig 1.3 Disk Image confirmed in the USB Directory

Then we do a little statistical summary analysis of the image source using `img_stat`



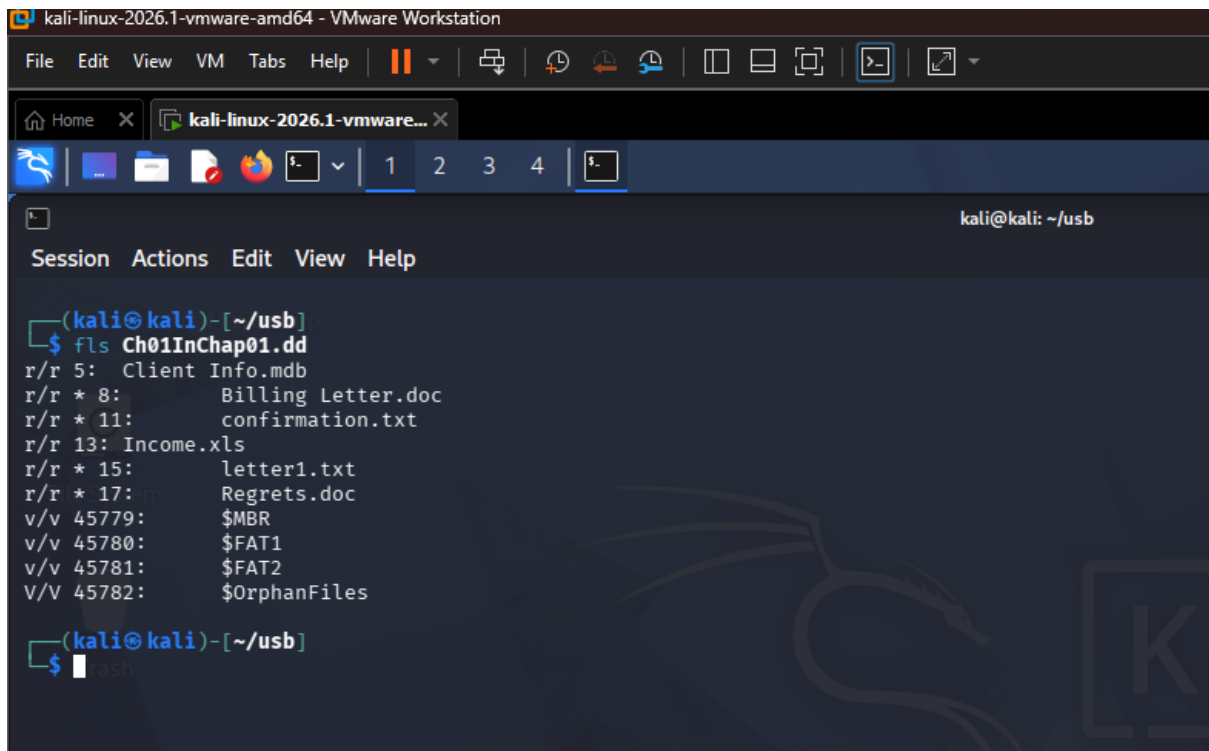
```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/usb
Session Actions Edit View Help
(kali@kali)-[~/usb]
$ ls Ch01InChap01.dd -l
-rw-rw-r-- 1 kali kali 1474560 Aug 28 13:23 Ch01InChap01.dd
(kali@kali)-[~/usb]
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
(kali@kali)-[~/usb]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
Image Type: raw
Size in bytes: 1474560
Sector size: 512
(kali@kali)-[~/usb]
$
```

Fig 1.4 Summary analysis of the Forensic Image



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/usb
Firefox ESR
Browse the World Wide Web
(kali@kali)-[~/usb]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: +3+M|
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
METADATA INFORMATION
Range: 2 - 45782
Root Directory: 2
CONTENT INFORMATION
```

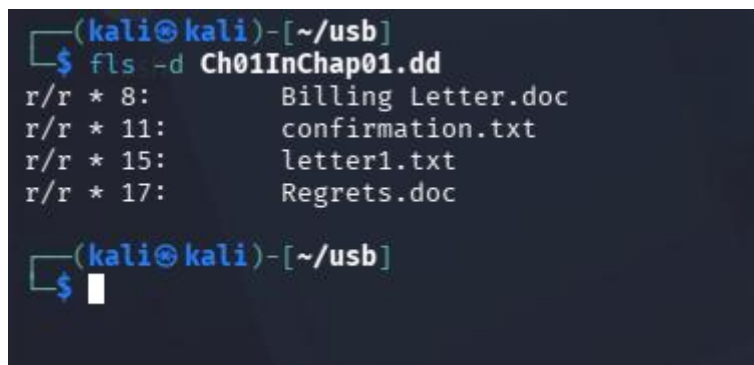
Fig 1.5 Showing Files system Details of the Source Image



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.1-vmware...
kali@kali: ~/usb
Session Actions Edit View Help
(kali@kali)-[~/usb]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
(kali@kali)-[~/usb]
$
```

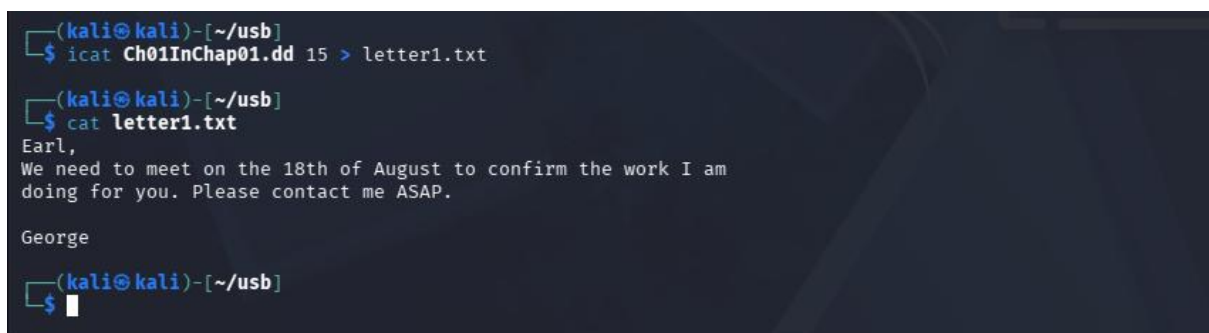
Fig 1.6 Showing the List of allocated and deleted File Names in the Directory

In Fig 1.6 above, here we see the INCOME.XLS File.



```
(kali@kali)-[~/usb]
$ fls -d Ch01InChap01.dd
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
(kali@kali)-[~/usb]
$
```

Fig 1.7 Displaying Deleted Entries only.



```
(kali@kali)-[~/usb]
$ icat Ch01InChap01.dd 15 > letter1.txt
(kali@kali)-[~/usb]
$ cat letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.
George
(kali@kali)-[~/usb]
$
```

Fig 1.8 Displaying Deleted Entries only.

```
(kali㉿kali)-[~/usb]
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (WAT)
Accessed:     2005-12-09 00:00:00 (WAT)
Created:      2005-12-09 06:59:07 (WAT)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

(kali㉿kali)-[~/usb]
$
```

Fig 1.9 Displaying the statistics about the files Metadata

This shows that this is a file with multiple sectors, Next we extract from the file sectors number

```
(kali㉿kali)-[~/usb]
$ blkcat Ch01InChap01.dd 312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

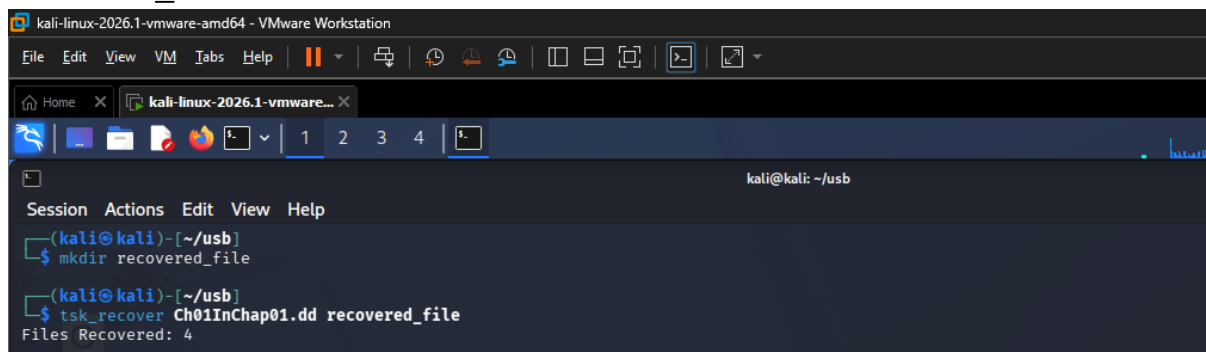
(kali㉿kali)-[~/usb]
$
```

Fig 1.10 Extracting the Content of the file from the Sectors Number using blkcat

Some of the commands used and their uses are:

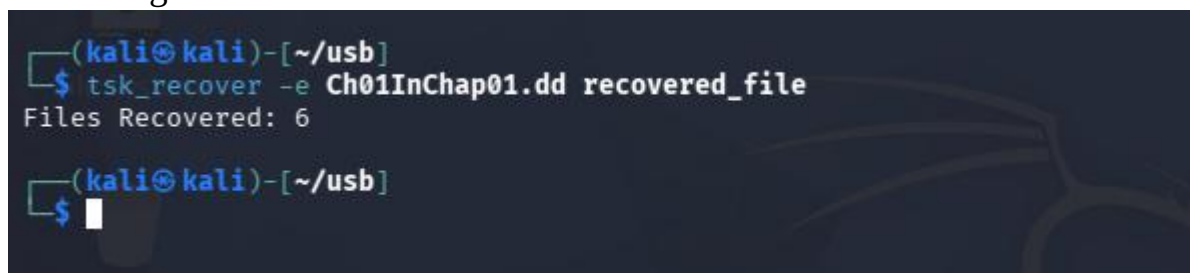
1. **Istat:** This examines metadata and timestamps of a specific file/inode.
2. **Fls:** Lists files and directories, including deleted files, in a disk image.
3. **Blkcat:** Displays the contents of a specific data block in the file system.
4. **Icat:** Recovers/extracts a specific file from a disk image using its inode.
5. **Fsstat:** Displays detailed information about the file system structure and metadata.

Using Tsk_Recover to recover deleted files, we first create a directory named `recovered_file`



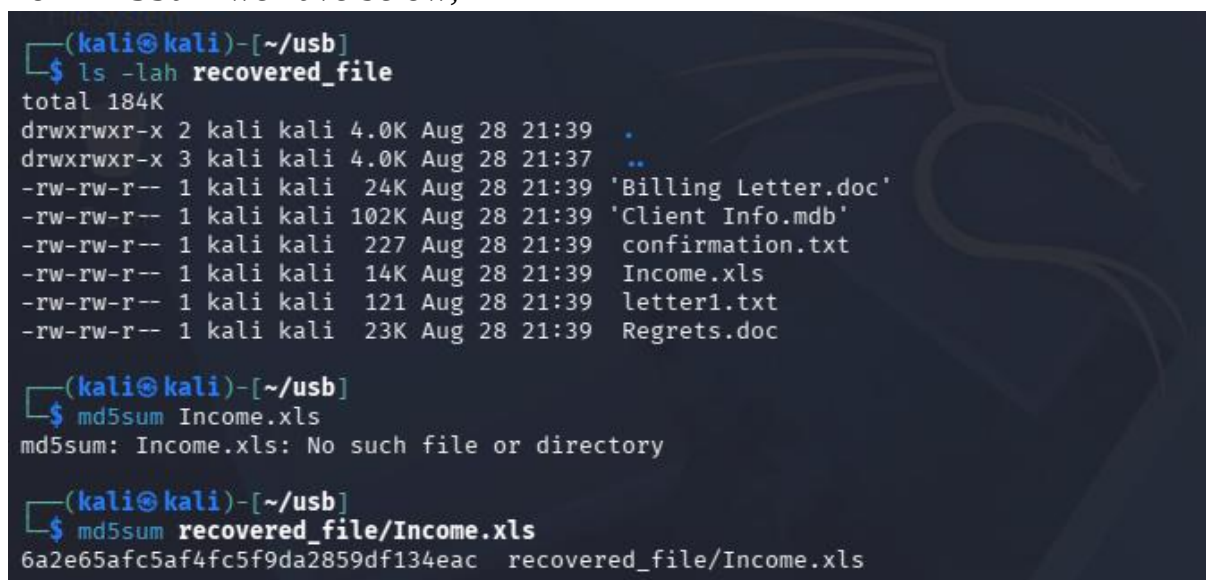
```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.1-vmware...
kali@kali: ~/usb
Session Actions Edit View Help
(kali@kali)-[~/usb]
$ mkdir recovered_file
(kali@kali)-[~/usb]
$ tsk_recover Ch01InChap01.dd recovered_file
Files Recovered: 4
```

When used without any option we recovered 4 files in total, but when used with the option `-e` it recovers both the allocated and the deleted files thereby recovering a total of 6 files.



```
(kali@kali)-[~/usb]
$ tsk_recover -e Ch01InChap01.dd recovered_file
Files Recovered: 6
(kali@kali)-[~/usb]
$
```

Now we compare the Hashes of the recovered File
For MD5Sum we have below,

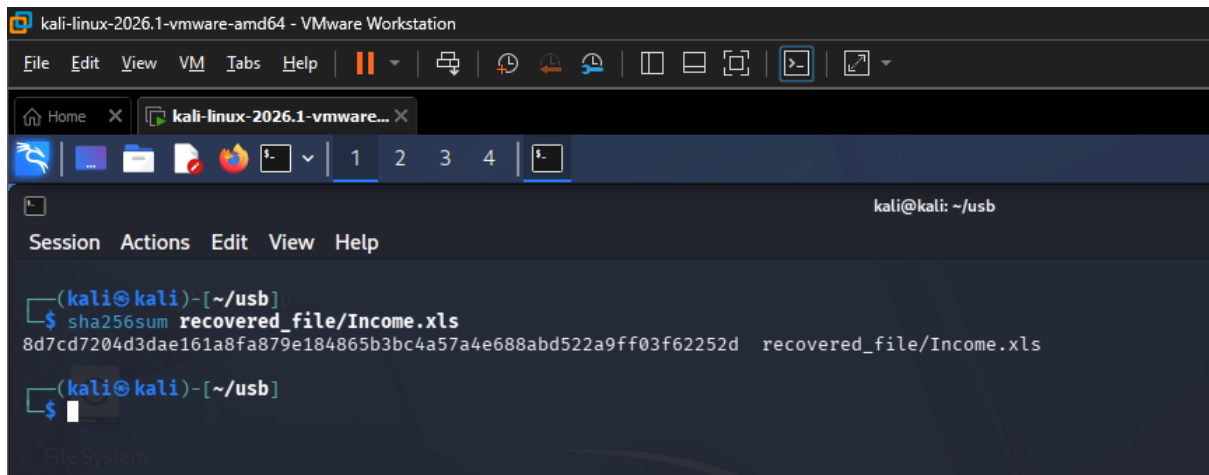


```
(kali@kali)-[~/usb]
$ ls -lah recovered_file
total 184K
drwxrwxr-x 2 kali kali 4.0K Aug 28 21:39 .
drwxrwxr-x 3 kali kali 4.0K Aug 28 21:37 ..
-rw-rw-r-- 1 kali kali 24K Aug 28 21:39 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 102K Aug 28 21:39 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 227 Aug 28 21:39 confirmation.txt
-rw-rw-r-- 1 kali kali 14K Aug 28 21:39 Income.xls
-rw-rw-r-- 1 kali kali 121 Aug 28 21:39 letter1.txt
-rw-rw-r-- 1 kali kali 23K Aug 28 21:39 Regrets.doc

(kali@kali)-[~/usb]
$ md5sum Income.xls
md5sum: Income.xls: No such file or directory

(kali@kali)-[~/usb]
$ md5sum recovered_file/Income.xls
6a2e65afc5af4fc5f9da2859df134eac recovered_file/Income.xls
```

And for the Sha256sum we have,



```
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2026.1-vmware... X
kali@kali: ~/usb
Session Actions Edit View Help
(kali@kali)-[~/usb]
$ sha256sum recovered_file/Income.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d  recovered_file/Income.xls
(kali@kali)-[~/usb]
$
```

In conclusion, the lab provided practical experience in preserving, analyzing, recovering, and validating digital evidence using Autopsy and The Sleuth Kit. It demonstrated how forensic tools and hash verification can ensure the integrity and reliability of recovered evidence during a digital investigation.